

Title: Early mobilization for acute heart failure: a systematic review and meta-analysis protocol

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M.K. is the guarantor. M.O. drafted the manuscript.

All authors contributed to the development of the selection criteria, the risk of bias assessment
strategy and data extraction criteria. M.O., Y.K., S.T., and T.F. developed the search strategy. M.O.,

Y.K., S.T., and T.F. provided statistical expertise. All authors read, provided feedback and approved the final manuscript.

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1.Introduction

Although the mortality rate of patients with heart failure has declining in recent years, they are older and have various comorbidities (multiple disabilities), and they often suffer from sarcopenia/frail due to poor physical function, which leads to repeated exacerbations and rehospitalization.^{1,2} The presence of overlapping multiple disabilities and sarcopenia/frail is a factor that prolongs hospitalization and reduces the rate of discharge home.^{3,4} In recent years, systematic reviews and RCTs have reported the benefits of early mobilization in patients on ventilators, stroke patients, and others.^{5,6} In ventilator patients, early mobilization has been reported to improve discharge rates from home, reduce ICU length of stay, and decrease complications related to hospitalization, such as pneumonia and delirium. In stroke patients, early mobilization has also been reported to improve physical functions. On the other hand, it has also been reported that early mobilization may have adverse effects, such as higher mortality at 1 year in patients with post-exacerbation chronic respiratory disease and worse functional independence at 3 months in patients with stroke, and its application should be treated with caution.^{7,8}

Early mobilization for inpatient acute heart failure has been applied in clinical practice with the same efficacy as in ventilator patients and stroke patients, but there is no consensus on the specific timing, efficacy, and safety of its initiation. For patients with heart failure, exercise therapy when their condition has stabilized has been shown to improve quality of life and reduce readmission rates.⁹ Although scoping reviews have been reported on early mobilization after acute myocardial infarction,¹⁰ no systematic review of early mobilization in patients with heart failure who are older^{11,12} and have a higher rate of multiple disabilities and physical deterioration¹³ than those with myocardial infarction has been reported, and its definition, efficacy, and safety have not been fully clarified. In heart failure, physical functions such as grip strength¹⁴ and activity of daily living (ADL)¹⁵ have been found to be associated with distant outcomes such as readmission rates and mortality, and early mobilization for acute heart failure, initiated to prevent physical conditioning and decline in ADLs, can contribute to these outcomes.

To the best of our knowledge, there are no scoping reviews on early mobilization for acute heart failure. The purpose of this study is to conduct a scoping review and systematic review of early mobilization for hospitalized heart failure to clarify its current definition and to evaluate its effectiveness and safety.

2.Research question

Part 1

Scoping review of early mobilization for heart failure (clinical definition)

The purpose of our scoping review is to clarify the current clinical definition of early mobilization

for acute heart failure.

(Table 1)

Part 2

Systematic review and meta-analysis of early mobilization for acute heart failure

Based on the scoping review above, we will design PICO and conduct a systematic review and meta-analysis to examine the efficacy and safety of early mobilization for acute heart failure.

(Table 2)

P: Hospitalized acute heart failure patients

I: Implement early mobilization (walking within three days or 72 hours of admission)

C: Do not implement early mobilization

O: Mortality, hospitalization days, readmission rates, ADL, mobility, quality of life, and adverse events (e.g., falls and changes in condition) associated with early mobilization

3.Method [Part 1]

We describe this protocol that follows the PRISMA extension for scoping reviews (PRISMA-ScR) statement.¹⁶ We also use the scoping review framework by The Joanna Briggs Institute (JBI)¹⁷ following the five stages approach as below.

Stage 1: Identifying the research question

We are interested in early mobilization for acute heart failure, and the research questions are as followed:

1. "What is the definition of early mobilization for acute heart failure?"
2. "When exactly is "early" in mobilization for acute heart failure?"
3. "What are the specific methods (implementation criteria, discontinuation criteria, frequency, time) of early mobilization for acute heart failure?"

Stage 2: Identifying relevant studies

We used the Population, Concept, and Context (PCC) framework¹⁷ to define the inclusion criteria as follows:

Participants

All publications that have been studied on acute heart failure will be included. The type of cardiovascular disease that causes acute heart failure is not relevant in this scoping review.

Concept

We will review the existing literature on early mobilization for acute heart failure regarding actual method, starting period, frequency and time.

Context

In this review, early mobilization is defined as "Activities such as sitting and walking that are started early after hospitalization". We will also include the study for early mobilization, early rehabilitation, and early physical therapy as defined by authors.

Additionally, we will only use MEDLINE via Ovid to search for Part 1. We will include randomized controlled trials and observational study, and not apply language or country restrictions. We will exclude abstracts, letters, case series study and case reports without controls. We will not exclude studies based on the observation period or publication year.

Stage 3: Study selection

We will select studies following the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) flow diagram.¹⁸ We will use the PCC framework and the process will be carried out independently by two researchers (M.O. and H.O.). The two authors will compare their lists and any differences in opinion will be resolved by discussion and, where this fails, through arbitration by a third researcher (M.K.).

Stage 4: Charting the data

Data extraction will be carried out by one researcher (M.O.) using standard data extraction forms including diagnosis, design, setting, age, sample size, definition of mobilization, number of starting days defined as "early", start and stop criteria, frequency, time, and dropout rate. Another one researcher (H.O.) confirms the data extraction. If necessary, we will contact the authors of these studies.

Stage 5: Collating, summarizing, and reporting the results

We will present the results of the search in the PRISMA flow diagram. We will organize the extracted data described above as qualitative synthesis (Table 1).

3. Method [Part 2]

3.1 Protocol

We used a systematic review protocol template ([dx.doi.org/10.17504/protocols.io.biqrkdv6](https://doi.org/10.17504/protocols.io.biqrkdv6)). We

followed the Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 for preparing this protocol.¹⁹ We will publish this protocol in protocols.io (<https://www.protocols.io/>).

3.2 Inclusion criteria of the articles for the review

3.2.1 Type of studies

We will include randomized controlled trials and observational studies that assess the effectiveness and safety of early mobilization for hospitalized acute heart failure patients. In observational studies, comparisons will be made between the patients who have received early mobilization (walking a few days of admission, early mobilization group) and those who have not (control group). We will not apply language or country restrictions. We will include all papers including published, unpublished articles, abstract of conference and letter.

We will exclude case series study and case reports without controls. We will not exclude studies based on the observation period or publication year.

3.2.2 Study participants

We will include hospitalized patients with acute heart failure. The details of inclusion/exclusion criteria are as follows.

Inclusion criteria:

The criteria for heart failure follow Framingham's diagnostic criteria or the authors' definition. 18 years of age or older.

Exclusion criteria:

Patients who underwent surgical procedures or treatment with catheterization after hospitalization will be excluded.

Patients who were unable to walk prior to admission should also be excluded.

3.2.3 Intervention

Implement early mobilization: patients should start walking within three days or 72 hours of admission, with or without assistance or aids (cane, etc.)^{20,21}.

The definition for the number of days of walking initiation will use the definitions extracted from the scoping review. Additionally, protocol-based interventions with progressively increasing rehabilitation stages, including walking, are also defined as interventions.

3.2.4 Control

Do not implement early mobilization: patients start walking after the above days of admission or do not walk.

3.3 Type of outcomes

3.3.1 Primary outcomes

1. Mortality

Definition: Include both cardiovascular and other deaths.

Period: During follow up period (use the longest follow-up period reported in the study)

2. Days of admission

Definition: Number of days spent in the acute care hospital where the patient was first admitted

Period: During follow up period

3. Readmission rate

Definition: For readmissions, it doesn't matter if it is the same hospital where the patient was initially admitted.

Period: During follow up period (use the longest follow-up period reported in the study)

3.3.2 Secondary outcomes

1. Activity of Daily Living (ADL) and mobility

Definition: Handling data that assesses ADL and mobility (regardless of the type of assessment chart)

Period: During follow up period (use the longest follow-up period reported in the study)

2. Quality of Life (QOL)

Definition: Deals with patient assessed quality of life data (regardless of the type of assessment chart)

Period: During follow up period (use the longest follow-up period reported in the study)

3. All adverse events

Definition: Definition of adverse events are set by original authors.

Incidence proportion of all adverse events

Period: During follow up period (use the longest follow-up period reported in the study)

3.4 Search method

3.4.1 Electronic search

We will search the following databases:

1. the Cochrane Central Register of Controlled Trials (CENTRAL);
2. MEDLINE via Ovid;
3. EMBASE via ProQuest Dialog;
3. CINAHL;
4. PEDro;

See Appendix 1, 2, 3, 4 and 5 for the search strategies.

3.4.2 Other resources

We will also search the following databases for ongoing or unpublished trials:

1. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP);
2. ClinicalTrials.gov; See Appendix 6, 7 for the search strategies.

We will check the reference lists of studies, including international guidelines²² as well as the reference lists of eligible studies and articles citing eligible studies. We will ask the authors of original studies for unpublished or additional data.

3.5 Data collection and analysis

3.5.1 Selection of the studies

Three groups of two pairs (six reviewers in total, M.O., H.O., A.I., M.S., Y.S., and R.S.) will screen titles and abstracts, followed by the assessment of the eligibility based on the full texts. We will contact original authors if relevant data is missing. Disagreements between the two reviewers will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter.

3.5.2 Data extraction and management

Two reviewers will perform independent data extraction of the included studies using standardized data collection form. We will use a pre-checked form using 10 randomly selected studies.

The form will include the information on study design, study population, interventions and outcomes (Table 1, Table 2). Any disagreements will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter. We will ask the authors of original studies for unclear points.

3.6 Assessment of risk of bias in included studies

Two reviewers will evaluate the risk of bias independently using the Risk of Bias 2 or ROBINS-I.^{23,24} Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer will be acting as an arbiter, if necessary.

3.7 Measures of treatment effects

We will pool the relative risk ratios/odds ratios and the 95% confidence intervals (CIs) for the following binary variables: mortality, readmission rate.

We will pool the mean differences and the 95% CIs for the following continuous variables: days of admission, ADL and mobility, QOL.

If several different scales have been used in the included studies, we will pool the effect estimates using standard mean differences (SMDs).

We will summarize adverse events based on the definition by the original article, but we will not perform meta-analysis.

For studies with a defined start date for walking or rehabilitation, we will include only observational studies in the meta-analysis. In addition, for studies of protocol-based interventions, we will include only randomized controlled trials in the meta-analysis. We should conduct the two meta-analyses separately.

3.8 Unit of analysis issues

Clustering at the level of the enrolled units in cluster randomized studies

In dealing with cluster-RCTs, for dichotomous data, we will apply the design effect and calculate effective sample size and number of events using the intracluster correlation coefficient (ICC) among each unit and the average cluster size, as described in Chapter 16.3.5 of the Cochrane Handbook.²⁵ If the ICC has not been reported, we will use the ICC of a similar study as a substitute. For continuous data, only the sample size will be reduced; means and standard deviation will remain unchanged.³

Randomized cross-over studies

We will consider only data from the first period.

Multiple comparisons

All intervention groups that are relevant to this review will be included.

3.9 Handling of missing data

We will ask not-presented data to the original authors.

3.9.1 Missing outcomes

We will perform the intention-to-treat (ITT) analysis for all dichotomous data as much as possible. For continuous data, we will not impute missing data based on the recommendation by Cochrane handbook.²⁵ We will perform meta-analysis about the available data in the original study.

3.9.2 Missing statistics

When original studies only report standard error or p-value, we will calculate the standard deviation based on the method by Altman.²⁶ If we don't know these values when we contact the authors, standard deviation will be calculated by confidence interval and t-value based on the method by Cochrane handbook²⁵, or validated method.²⁶ Validity of these methods will be analyzed by sensitivity analysis.

3.10 Assessment of heterogeneity

We will evaluate the statistical heterogeneity by visual inspection of the forest plots and calculating the I² statistic (I² values of 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity). When there is substantial heterogeneity (I² > 50%), we will assess the reason for the heterogeneity. Cochrane Chi² test (Q-test) will be performed for I² statistics, and P value less than 0.10 will be defined as statistically significant.

3.11 Assessment of reporting bias

We will search the clinical trial registry system (ClinicalTrials.gov and ICTRP) and will perform extensive literature search for unpublished trials. We will assess the potential publication bias by visual inspection of the funnel plot. Egger test will be performed as well. We will not conduct the test when we find less than 10 trials or trials which have similar sample sizes. We will assess the potential publication bias by visual inspection of the funnel plot.

3.12 Meta-analysis

Meta-analysis will be performed using Review Manager software (RevMan 5.4). We will use a random-effects model.

3.13 Subgroup analysis

To elucidate the influence of effect modifiers on results, we will evaluate the subgroup analyses of the primary outcomes on the following factors when sufficient data are available.

1. (For participants) Over 65 years old and under
2. (For intervention) Early mobilization (walking) up to and after the second day of hospitalization

3.14 Sensitivity analysis

We will undertake the following sensitivity analyses for the primary outcomes to assess whether the results of the review are robust to the decisions made during the review process.

1. Exclusion of studies using imputed statistics.

2. Missing participants: verify the robustness of the results by seeking informative missingness odds ratios.²⁷

3. Only the participants who complete the study with complete data

4. Inclusion of studies that have diagnosed heart failure with Framingham's diagnostic criteria or the authors' definition. Patients with heart failure who do not meet these criteria should be excluded.

5. For the main analysis, the definition extracted from the scoping review will be used for the criterion of early mobilization (the number of days starting from admission), but sensitivity analysis will be performed for the other days.

4. Summary of findings table

Summary of findings table will be made for the following outcome based on the Cochrane handbook.²⁵

We will include grading to evaluate the quality of evidence based on the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach for each Summary of findings table.²⁸

1. Mortality
2. Hospital length of stay
3. Readmission ratio
4. ADL and mobility
5. QOL
6. All adverse events

5. Conflict of Interest

The authors declare no conflicts of interests.

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Appendix 1: CENTRAL search strategy

1. (((((((([mh "heart failure"]) OR ("heart failure":ti,ab)) OR ("heart insufficiency":ti,ab)) OR ("cardiac failure":ti,ab)) OR ("cardiac insufficiency":ti,ab)) OR ("myocardial failure":ti,ab)) OR ("myocardial insufficiency":ti,ab)) OR (HFNEF:ti,ab)) OR (HFPEF:ti,ab)) OR (HFREF:ti,ab)) OR ("HF NEF":ti,ab)) OR ("HF PEF":ti,ab)) OR ("HF REF":ti,ab)) AND (((((((early:ti,ab) OR (immediate*:ti,ab)) OR (accelerat*:ti,ab)) OR ("fast track":ti,ab)) OR (timing:ti,ab)) OR (rapid:ti,ab)) AND (([mh "early ambulation"]) OR ([mh "exercise therapy"]))) OR (((((((early:ti,ab) OR (immediate*:ti,ab)) OR (accelerat*:ti,ab)) OR ("fast track":ti,ab)) OR (timing:ti,ab)) OR (rapid:ti,ab)) NEXT (((((((mobilization:ti,ab) OR (mobilisation:ti,ab)) OR (mobility:ti,ab)) OR (exercise:ti,ab)) OR ("physical therapy":ti,ab)) OR (physiotherapy:ti,ab)) OR (ambulation:ti,ab)) OR (rehabilitation:ti,ab)) OR ("cardiac rehabilitation":ti,ab))))))

Appendix 2: MEDLINE (via Ovid) search strategy

1. (((((((((((exp heart failure/) OR (heart failure.ti,ab.)) OR (heart insufficiency.ti,ab.)) OR (cardiac failure.ti,ab.)) OR (cardiac insufficiency.ti,ab.)) OR (myocardial failure.ti,ab.)) OR (myocardial insufficiency.ti,ab.)) OR (HFNEF.ti,ab.)) OR (HFPEF.ti,ab.)) OR (HFREF.ti,ab.)) OR (HF NEF.ti,ab.)) OR (HF PEF.ti,ab.)) OR (HF REF.ti,ab.)) AND (((((((early.ti,ab.) OR (immediate*.ti,ab.)) OR (accelerat*.ti,ab.)) OR (fast track.ti,ab.)) OR (timing.ti,ab.)) OR (rapid.ti,ab.)) AND ((exp early ambulation/) OR (exp exercise therapy/))) OR (((((((early.ti,ab.) OR (immediate*.ti,ab.)) OR (accelerat*.ti,ab.)) OR (fast track.ti,ab.)) OR (timing.ti,ab.)) OR (rapid.ti,ab.)) ADJ (((((((mobilization.ti,ab.) OR (mobilisation.ti,ab.)) OR (mobility.ti,ab.)) OR (exercise.ti,ab.)) OR (physical therapy.ti,ab.)) OR (physiotherapy.ti,ab.)) OR (ambulation.ti,ab.)) OR (rehabilitation.ti,ab.)) OR (cardiac rehabilitation.ti,ab.))))))

Appendix 3: EMBASE (via ProQuest Dialog) search strategy

- S1 EMB.EXACT.EXPLODE("acute heart failure")
- S2 ab("heart failure") OR ti("heart failure")
- S3 ab("heart insufficiency") OR ti("heart insufficiency")
- S4 ab("cardiac failure") OR ti("cardiac failure")
- S5 ab("cardiac insufficiency") OR ti("cardiac insufficiency")
- S6 ab("myocardial failure") OR ti("myocardial failure")
- S7 ab(myocardial insufficiency) OR ti("myocardial insufficiency")
- S8 ab("HFNEF") OR ti("HFNEF")
- S9 ab("HFPEF") ` OR ti("HFPEF")

S10 ab("HFREF") OR ti("HFREF")
 S11 ab(HF NEF") OR ti("HF NEF")
 S12 ab("HF PEF") OR ti("HF PEF")
 S13 ab("HF REF") OR ti("HF REF")
 S14 S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR
 S13
 S15 EMB.EXACT.EXPLODE("mobilization")
 S16 EMB.EXACT.EXPLODE("exercise")
 S17 ab("mobilization") OR ti("mobilization")
 S18 ab("mobilisation") OR ti("mobilisation")
 S19 ab("mobility") OR ti("mobility")
 S20 ab("exercise") OR ti("exercise")
 S21 ab("physical therapy") OR ti("physical therapy")
 S22 ab("physiotherapy") OR ti("physiotherapy")
 S23 ab("ambulation") OR ti("ambulation")
 S24 ab("rehabilitation") OR ti("rehabilitation")
 S25 ab("cardiac rehabilitation") OR ti("cardiac rehabilitation")
 S26 ab(early) OR ti(early)
 S27 ab(immediate*) OR ti(immediate*)
 S28 ab(accelerat*) OR ti(accelerat*)
 S29 ab("fast track") OR ti("fast track")
 S30 ab(timing) OR ti(timing)
 S31 ab(rapid) OR ti(rapid)
 S32 S26 OR S27 OR S28 OR S29 OR S30 OR S31
 S33 S15 OR S16
 S34 S32 AND S33
 S35 S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25
 S36 S32 NEAR/ S35
 S37 S14 AND (S34 OR S36)

Appendix 4: CHNAHL search strategy

1. (((((((((((((MH "heart failure+") OR ((TI "heart failure" OR AB "heart failure")) OR ((TI "heart insufficiency" OR AB "heart insufficiency")) OR ((TI "cardiac failure" OR AB "cardiac failure")) OR ((TI "cardiac insufficiency" OR AB "cardiac insufficiency")) OR ((TI "myocardial failure" OR AB "myocardial failure")) OR ((TI "myocardial insufficiency" OR AB "myocardial insufficiency")) OR ((TI HFNEF OR AB HFNEF)) OR ((TI HFPEF OR

AB HFPEF))) OR ((TI HFREF OR AB HFREF))) OR ((TI "HF NEF" OR AB "HF NEF")))
 OR ((TI "HF PEF" OR AB "HF PEF")) OR ((TI "HF REF" OR AB "HF REF")) AND
 (((((((((TI early OR AB early)) OR ((TI immediate* OR AB immediate*)) OR ((TI accelerat*
 OR AB accelerat*)) OR ((TI "fast track" OR AB "fast track")) OR ((TI timing OR AB
 timing))) OR ((TI rapid OR AB rapid))) AND (((MH "early ambulation+") OR ((MH
 "exercise therapy+")))) OR (((((((((TI early OR AB early)) OR ((TI immediate* OR AB
 immediate*)) OR ((TI accelerat* OR AB accelerat*)) OR ((TI "fast track" OR AB "fast
 track")) OR ((TI timing OR AB timing))) OR ((TI rapid OR AB rapid))) W1 (((((((((TI
 mobilization OR AB mobilization)) OR ((TI mobilisation OR AB mobilisation))) OR ((TI
 mobility OR AB mobility))) OR ((TI exercise OR AB exercise))) OR ((TI "physical therapy"
 OR AB "physical therapy")) OR ((TI physiotherapy OR AB physiotherapy))) OR ((TI
 ambulation OR AB ambulation))) OR ((TI rehabilitation OR AB rehabilitation))) OR ((TI
 "cardiac rehabilitation" OR AB "cardiac rehabilitation"))))

Appendix 5: PEDro search strategy

1. acute heart failure

Appendix 6: ICTRP search strategy

1. (acute heart failure) AND ("exercise therapy" OR mobilization OR mobilisation OR mobility
 OR exercise OR "physical therapy" OR physiotherapy OR ambulation OR rehabilitation OR
 "cardiac rehabilitation")

Appendix 7: ClinicalTrials.gov search strategy

Condition or disease: acute heart failure

Intervention: "exercise therapy" OR mobilization OR mobilisation OR mobility OR exercise OR
 "physical therapy" OR physiotherapy OR ambulation OR rehabilitation OR "cardiac
 rehabilitation"

Table 1 Details and starting days of early mobilization for acute heart failure in each study

Author, Year, Country	Design, Setting	Age, Sample size	Definition of mobilization (walking, standing, etc)	Number of starting days defined as "early"	Start and stop criteria	Frequency, time	Dropout rate

Table 2

Author, Year, Country	Design, Setting	Age, Sample size	Intervention	Controls	Outcomes